ANNEXURE-I

Computer Organization and Assembly Language (COAL)	
Course Code	CS-2165
Semester	Fall 2024
Cr. Hrs.	4 (3+1)
Section (s)	COAL 3A, 3C, 3P

Instructor Contact Information

Instructor	Mr.Muhammad Yasin Nasir
Email	yasin.nasir@cs.uol.edu.pk
Cell No.	
Zoom ID	
Office	Room No 109 First Floor CS Building
Student Consultation Hours	i) Teacher will be available during the office hours for student consultation as per following schedule: **Monday:** 12:00 PM to 2:00 PM, **Tuesday:** 2:00 PM to 3:00 PM, **Thursday:** 10:00 AM to 12:00 PM

Class Details

Section	Class Type	Class Title	Days & Times	Class Room Location	Class Dates

Communication with Instructor	i) Please use email rather than telephone voice mail for messages. ii) Please keep emails short and focused, and use a clear subject line beginning with "COMP 110 Question". iii) Instructor will generally respond within 24 hours (during the academic days). iv) Always include your **name**, **course**, and **UOL email address** in your messages to instructor. If you send an email from some address **other than UOL email address** like mr@abcd.com; instructor would not be able to recognize that you are a student of University.

Pre-Requisites: (Note: Instructor should define it in coordination with HOD/Faculty Committee)

Prerequisites:	
Corequisite	

Medium of Instruction

Medium of Instruction	Instructor will deliver class lectures, conduct discussions and engage in all academic activities exclusively in English. Moreover, the students are required to use English as mode of communication for class room discussions, group discussions, vivas/presentations, group projects etc.

Course Material Availability on SLATE/LMS:

SLATE/LMS	I will use LMS/Slate to keep in touch with my classes. In-class discussion problems, homework solutions and supplemental materials, and scores for exams, attendance, and assignments will be posted to LMS/SAP. You should be certain that you are able to log in to LMS/SAP, and that you check the class page on LMS and your University email account regularly.
	Course material is available on LMS: https://slate.uol.edu.pk/login/index.php
	Grades will be posted on LMS/SAP: https://slate.uol.edu.pk/login/index.php

Required Text

Text Book 1	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)
Text Book 2	Computer System Architecture 3rd Edition by Morris Manno
Other Material	1. 80X86 IBM PC and Compatible Computers: Assembly Language, Design, and Interfacing", Volumes I & II (5th Edition) 2010, Pearson by Muhammad Ali 2. Mazidi The Intel Microprocessors 8th Edition, By Barry B. Brey

Software/ Other Tools/Resources

Emulator 8086	8086 Microprocessor Emulator Setup You must install the 8086 Microprocessor Emulator to write and run COAL programs for this class. Installation Instructions: 1. Go to https://emu8086-microprocessor-emulator.en.softonic.com/download 2. Click the "Download" button. 3. Follow the installation prompts to complete the setup.
DOSBOX MASM	You can also install DOSBox and MASM (Microsoft Macro Assembler) to write and run Assembly language programs for this class. **Installation Instructions:** Step 1: Install DOSBox 1. Go to https://drive.google.com/file/d/0B36ww40izFp6dkVWOVk0c1VnOFE/view?resourcekey=0-C9jAxlBJyoIEX7f8Kz5wzg 2. Click the "Download" button. 3. Follow the installation prompts to complete the setup for DOSBOX0.74. Step 2: MP/MASM Folder 1. Create a new folder for your MASM projects (e.g., C:\MASM or C:\MP).

	2. Copy the MASM/MP files to this folder. Running MASM in DOSBox
Other Tools/Softwares	i) VS Code with assembly language extension

Goals: (Note: Instructor should define it in coordination with HOD/Faculty Committee)

Sr. No	Goals
1	Develop a deep understanding of how computers function at the hardware level, including knowledge of key components like the CPU, memory, input/output devices, and how they interact to perform tasks.
2	Learn about instruction set architectures (ISAs) and how assembly language maps directly to the hardware through the CPU's machine language instructions.
3	Gain proficiency in writing, assembling, and executing programs in assembly language for a specific processor (e.g., x86, MIPS).
4	Understand low-level data operations, such as memory addressing, data movement, arithmetic/logic operations, and control flow at the instruction level.
5	Explore how assembly language can be used to optimize program performance, as it provides fine control over the hardware, including register usage and instruction selection.
6	Work on practical lab exercises to implement and debug assembly programs, as well as simulate hardware operations.
7	Learn how high-level languages like C and Java are translated into assembly code and understand the role of compilers and linkers in this process.

Course Objectives: (Note: Instructor should define it in coordination with HOD/Faculty Committee)

Sr. No	Objectives
1	Able to identify distinguishing features of Intel family members ISA.
2	Able to Understand functions of modern memory & I/O systems and interface them to the microprocessors
3	Develop software to interface Intel microprocessors with memory and IO.
4	Analyze, design and implement practical systems of up to average complexity within a team.
5	Appreciate design issues related to multi-core processor systems

COURSE REQUIREMENTS (Subject to Change)

(Note: Instructor should define it in coordination with HOD/Faculty Committee)

A. Class Attendance:

You are expected to attend class and participate by asking questions, answering questions and contributing to topical discussions. You are expected to arrive prior to the beginning of class. Class members will be randomly selected to answer questions and/or help with the solution of exercises.

If you do miss a class, it is your responsibility to ensure that you understand the material covered and the announcements made in the class you missed. You may not use office hours to have the professor explain the material if you missed the class when it was discussed.

B. Class Participation:

The following factors will be considered in evaluating class participation:

(1) Attend class sessions and contribute to a positive learning environment,
(2) Ask thoughtful questions,
(3) Participate in discussion,
(4) Prepare for reading and assignments conscientiously.

A positive learning environment results when you demonstrate respect for other students and the instructor, are courteous and attentive, assist others in learning, attend class regularly, and arrive in class on time.

C. Homework/Assignment:

Doing the homework is essential for success in this course. Throughout the semester, at least **4 Homework/Assignments (5 marks/points each) , 2 before Mid-Term** and **2 after Mid-Term** will be given.

Homework/Assignment assigned will be discussed in class when time permits; but you may not understand it completely unless you have already attempted the work. Programming is a discipline that cannot be learned merely by watching; it is learned only by doing. By attempting each homework problem prior to the class discussion, you can maximize your learning experience and will be in a position to ask appropriate questions and identify areas where you need help. Accordingly, homework should be done as instructed before coming to each class.

Incomplete work (or cheating on an assignment) cannot be accepted. **Late homework cannot be accepted** (as the solution will be posted right after the collection of the assigned homework).

D. Quizzes

Throughout the semester, at least **4 quizzes (5 marks/points each)**, **2 before Mid-Term** and **2 after Mid-Term** will be given in class on material covered in the previous week(s) to ensure that you are keeping current. **There are no make-up quizzes**. If you are not in class on the day of a quiz, you will receive a zero.

E. Exams:

There will be **two (2) exams for the semester**. All exams are "closed book." Students will not be given any extended time if arriving late on that day.

Exams will be a combination of objective questions (e.g., multiple choices, short answer) and numerical problems. To receive credit for numerical answers, adequate supporting computations (and also explanations if necessary) must be provided.
No makeup exam will be given and the exam will not be administered on any other day!

EVALUATION & GRADING :

(Note: Teacher should modify it as per University/Department Policy in Consultation with HOD/Faculty Committee)

The course grades will be determined by the following:

Components	Marks	Weight
Mid-Term	40	20%
Lab	100	20 %
Final-Exam	40	40 %
Quizzes (4*5 points)	80	10%
Home Work/Assignment (4*5 points)	80	10%
Total		100%

Your Responsibilities

Sr. No.	Major Responsibilities
1	**Arrive on time and do not leave before the end of the class period.**
2	**Refrain from causing other distractions** (ringing cell phones, talking while others have the floor, etc.). Disruptive students will be asked to leave the class
3	**Treat everyone with respect**
4	**Be responsible for all assigned materials plus everything covered in class**. If you missed classes, make arrangements with another student to collect handouts and to update you on classes you missed

ACADEMIC HONESTY STATEMENT

Academic dishonesty, which includes cheating, fabrication, facilitation of academic dishonesty, and plagiarism, is a serious academic offense. A grade of "F" shall be assigned to any student who engages in academic dishonesty in this class, and formal disciplinary action shall be taken. (Plagiarism in any assignment or cheating in the examinations will result in a grade of F in the entire course).

THIS SYLLABUS CONSTITUTES A CONTRACT BETWEEN THE STUDENT AND THE FACULTY OF IT (FIT). THE TERMS AND CONDITIONS CONTAINED IN THIS CONTRACT ARE DEEMED TO BE ACCEPTED BY EACH STUDENT WHO REMAINS IN THIS COURSE AFTER THE OFFICIAL DROP DATE.

Home Work/Assignment and Quiz Details (subject to change)

Home Work (HW)/Assignment					Quiz				
Homework/ Assignment (on LMS)	**Week No**	**HW/Assignment Submission by Student on LMS**	**Solution Posting of HW/Assignment By Teacher on LMS**	**Result Submission Date**	**Quiz (on LMS)**	**Week No**	**Submission of Quiz by Student on LMS**	**Solution Posting of Quiz by Teacher on LMS**	**Result Submission date**
Home Work 1	Week 2-3	Within the Time Line Assigned by Teacher	Immediately after the assignment submission dead line	Within 7days	**Quiz 1**	Week 3-4	Within the Time Line Assigned by Teacher	Immediately after the quiz submission dead line	Within 7days
Home Work 2	Week 5-6	Same as Above	Same as Above	Within 7days	**Quiz 2**	Week 6-7	Same as Above	Same as Above	Within 7days
Home Work 3	Week 10-11	Same as Above	Same as Above	Within 7days	**Quiz 3**	Week 11-12	Same as Above	Same as Above	Within 7days
Home Work 4	Week 12-13	Same as Above	Same as Above	Within 7days	**Quiz 4**	Week 13-14	Same as Above	Same as Above	Within 7days
Reading Assignments	Every Week								

Note : ***This is the minimum number of Home Work/Assignments and Quizzes . The Instructor should encourage to add more Home Work/Assignments and Quizzes.***

Submitting Work:

1. Most assignments will be submitted electronically via Slate/LMS.
2. Please do not email submissions until and unless necessary due some justified reason.

Course Contents and Tentative Class Schedule Table-I (subject to change)

Week No	Dates	Topics	Book Name	Chapter No	Exam/Quiz/Assignment	
					Category	Time Line for Result Submission
1		Introduction to Computer Organization and Assembly Language	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	1	NA	NA
		History & Evolution of Intel Microprocessor and Assembly Language	Same as Above	3	NA	NA
		Applications and Advantages of Assembly Language	Same as Above	3	NA	NA
		Organization of Intel 80x86 Microprocessor	Same as Above	4	NA	NA
		Architecture: Memory Hierarchy, Data Bus, Control Bus, Address Bus	Same as Above	4	NA	NA
2		Registers and their categories	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	4	NA	NA
		Introduction to Program Segments	Same as Above	4	Home Work/ Assignment 1 (Week 2/Week3)	Within 7 Days
		Memory Segmentation: Logical Address and Physical Address	Same as Above	3	NA	NA

Week No	Dates	Topics	Book Name	Chapter No	Exam/Quiz/Assignment	
					Category	Time Line for Result Submission
3		Directives and A Sample Program	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	4	NA	NA
		How to write and execute sample program using MASM DOS Interrupt 21H, Single Character Input, Output, String Output, New Line	Same as Above	4	Quiz 1 (Week 3/Week4)	Within 7 Days
		Data Definition: DB, DW, DD, DQ, DT Different variants of MOV and XCHG instruction	Same as Above	5	NA	NA
		Declare A Variable, Arrays, DUP, EQU, ORG Variables Naming Convention 80x86 Addressing Modes	Same as Above	5	NA	NA
4		Arithmetic Instructions: ADD, SUB, INC, DEC, NEG Signed and Unsigned Operations	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	5	NA	NA
		Logical Instructions: AND, OR, XOR, NOT, TEST	Same as Above	7	NA	NA
5		Flags CF, ZF, SF, PF, AF, OF Effect of Arithmetic and Logical Instructions on Flags	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	6	NA	NA
		Signed and Unsigned Jump, Conditional and Unconditional, Single flag and Multi-flag Jumps. (JG, JNG, JL, JNL, JA, JNA, JAE, JNAE etc.)	Same as Above	6	Home Work/ Assignment 2 (Week 5/Week6)	Within 7 Days
6		Control Transfer Functions	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	6	Quiz 2 (Week 6/Week7)	Within 7 Days
		Loop Structures	Same as Above	6		
7		Rotate Instructions, ROL, ROR	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	7	NA	NA
		RCL, RCR	Same as Above	7	NA	NA
		Shift Instructions: SHL, SHR, SAR, SAL		7	NA	NA

Week No	Dates	Topics	Book Name	Chapter No	Exam/Quiz/Assignment	
					Category	Time Line for Result Submission
		Stack PUSH, POP, CALL, RET, PUSHF, POPF	Same as Above	8	NA	NA
8		Procedures: Input Binary, Hexadecimal, Decimal Output Binary, Hexadecimal, Decimal	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	8	NA	NA
		Procedures: Binary ⇋ Hexadecimal Decimal ⇋ Hexadecimal Binary ⇋ Decimal	Same as Above	8	NA	NA
9		**Mid Term Exam Week**			Midterm	Within 5 Days
10		MUL Instruction	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	9	NA	NA
		DIV instruction	Same as Above	9	Home Work/ Assignment 3 (Week10/Week11)	Within 7 Days
11		XLAT Instruction String Operations: MOVSB/W, LOADSB/W, STOSB/W	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	10,11	NA	NA
		String Operations: SCASB/W, CMPSB/W	Same as Above	11	Quiz 3 (Week 11/ Week12)	Within 7 Days
12		INT 16H Keyboard Programming	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	12	Home Work/ Assignment 4 (Week12/Week13)	Within 7 Days
		BIOS INT 10H Programming, Set Cursor position and get cursor position	Same as Above	15	NA	NA
13		File Operations: Reading a File, Writing a File	Asssembly Language Programming and Organization IBM PC 1st Edition by Ytha Yu (Author), Charles Marut (Author)	19	Quiz 4 (Week 13/Week14)	Within 7 Days
		Organization of Basic Computer	Computer System Architecture 3rd Edition by Morris Manno	5	NA	NA

<table>
<tr><th rowspan="2">Week No</th><th rowspan="2">Dates</th><th rowspan="2">Topics</th><th rowspan="2">Book Name</th><th rowspan="2">Chapter No</th><th colspan="2">Exam/Quiz/Assignment</th></tr>
<tr><th>Category</th><th>Time Line for Result Submission</th></tr>
<tr><td rowspan="2">14</td><td rowspan="2"></td><td>Instruction Set Architecture:
Memory Reference
Register Reference
I/O Instructions</td><td>Computer System Architecture 3rd Edition by Morris Manno</td><td>5</td><td>NA</td><td>NA</td></tr>
<tr><td>Instruction Cycle:
Fetch, Decode, Execute</td><td>Computer System Architecture 3rd Edition by Morris Manno</td><td>5</td><td>NA</td><td>NA</td></tr>
<tr><td rowspan="2">15</td><td rowspan="2"></td><td>Register Transfer Language:
Register Transfer
Bus Transfer
Memory Transfer</td><td>Computer System Architecture 3rd Edition by Morris Manno</td><td>4</td><td>NA</td><td>NA</td></tr>
<tr><td>Logic Microoperations</td><td>Same as Above</td><td>4</td><td>NA</td><td>NA</td></tr>
<tr><td rowspan="2">16</td><td rowspan="2"></td><td>Arithmetic Microoperations</td><td>Computer System Architecture 3rd Edition by Morris Manno</td><td>4</td><td>NA</td><td>NA</td></tr>
<tr><td>Design of ALU for Basic Computer</td><td>Same as Above</td><td>5</td><td>NA</td><td>NA</td></tr>
<tr><td>17</td><td></td><td>Final Exam Week</td><td></td><td></td><td rowspan="2">Final Term</td><td rowspan="2">Within 3 Days</td></tr>
<tr><td>18</td><td></td><td>Final Exam Week</td><td></td><td></td></tr>
</table>